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| <b>9</b><br><br><b>Designing for maintenance and the cleaner environment</b> | 9.1 | <p>Characteristics, application, advantages and disadvantages of 'cleaner' design and technology – a product's life cycle in relation to the following sustainable development issues:</p> <ul style="list-style-type: none"> <li>a) material selection – source, quantity, quality, range, recyclability, biodegradability</li> <li>b) manufacture – minimising energy use, simplification of processes, achieving optimum use of materials and components, giving consideration to material form, cost and scale of production</li> <li>c) distribution – efficient use of packaging, reduction of transport, alternatives to fossil fuels</li> <li>d) use – repair versus replacement, energy efficiency, efficiency ratings</li> <li>e) repair and maintenance – standardisation, modular construction, bought in parts</li> <li>f) end of life – design for disassembly, recovered material collection, sorting and re-processing methods, energy recovery, environmental implications of disposal to landfill.</li> </ul> |
|                                                                              | 9.2 | <p>The wider issues of using cleaner technologies:</p> <ul style="list-style-type: none"> <li>a) cost implications to the consumer and manufacturer</li> <li>b) sustainability – designing without jeopardising the potential for people in the future to meet their needs.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| Topic | What students need to learn: |
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